

ITA0608 - Machine Learning

List of Lab Exercises

Confusion Matrix

Aim

To implement a Decision Tree Classifier using Python and evaluate its performance by generating a Confusion Matrix.

Algorithm

1. Import the required libraries.
2. Load the dataset from the `trainingdata.csv` file.
3. Separate the features (X) and target (y).
4. Convert categorical data into numerical values if necessary.
5. Split the dataset into training and testing sets.
6. Train the Decision Tree classifier using the training data.
7. Predict the class labels for the test data.
8. Generate the Confusion Matrix.
9. Display the Confusion Matrix and classification accuracy.

Program

```
import pandas as pd

from sklearn.tree import DecisionTreeClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import ConfusionMatrixDisplay

import matplotlib.pyplot as plt

data = pd.read_excel("student_data.xlsx")

X = data[['Hours_Studied', 'Attendance']]

y = data['Pass']
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

ConfusionMatrixDisplay.from_predictions(y_test, y_pred)

plt.show()

```

Output

```

import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Load dataset
data = pd.read_excel("student_data.xlsx")

X = data[['Hours_Studied', 'Attendance']]
y = data['Pass']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Display Confusion Matrix
ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.show()

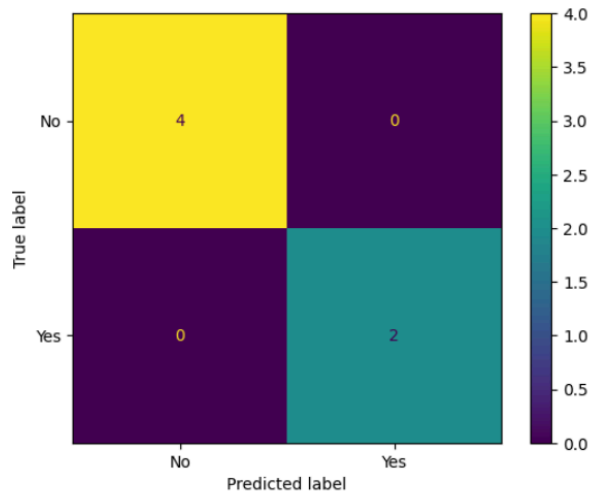
```



```
# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Display Confusion Matrix
ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.show()
```



Result

The Decision Tree Classifier was successfully implemented using Python. The model was evaluated using a Confusion Matrix, and the classification accuracy was found to be 100% (1.0) for the given dataset.